AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 31 · STUDENT LESSON

Permissions and Human Review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Before you start

Week 30: a tool action has a defined permission scope.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of least privilege and tool scopes. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of **least privilege** and tool scopes.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Agents and Tools. Give an example and a mistake that the idea helps you avoid.

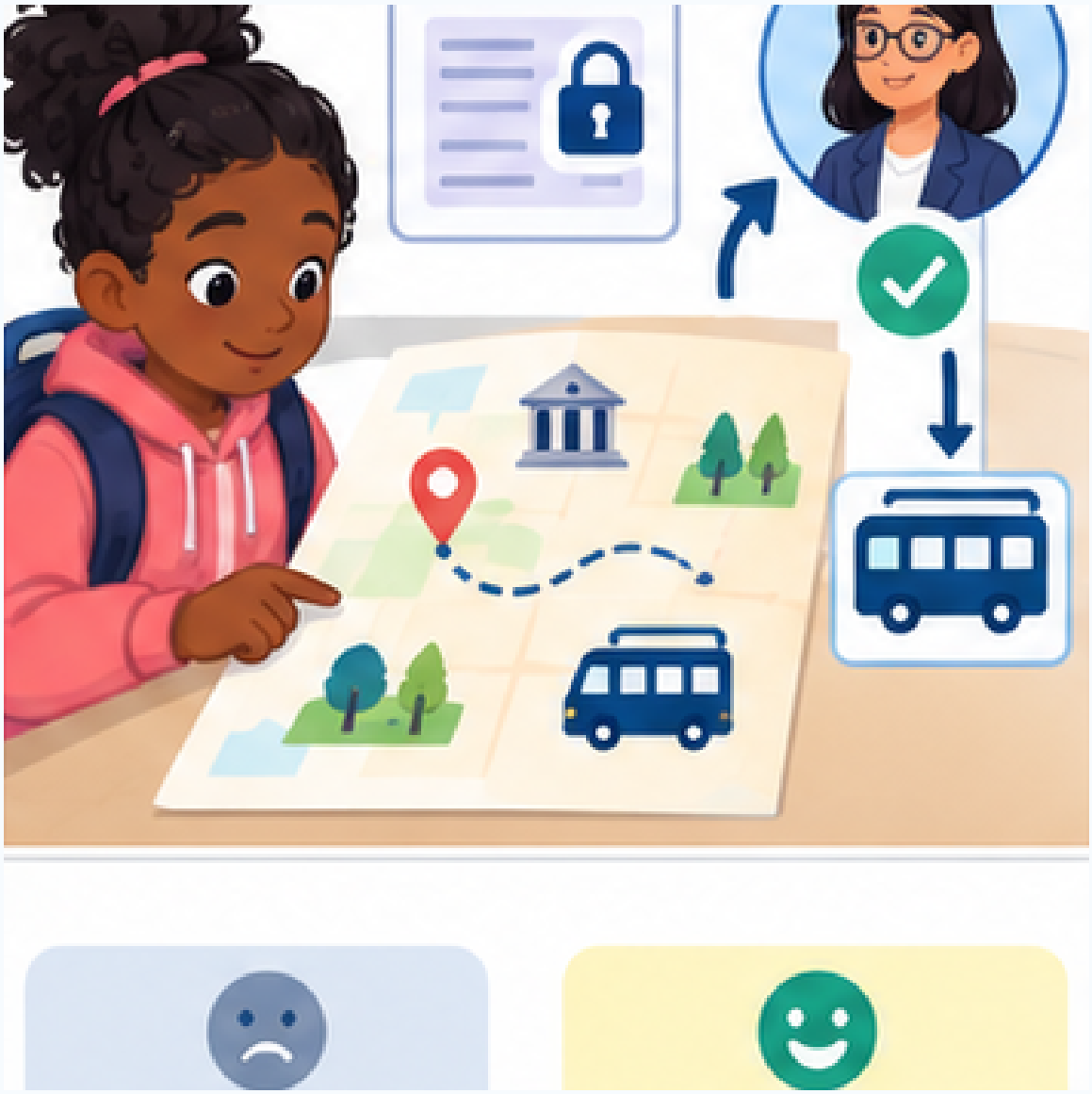
Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Text as Pieces. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 31: **Permission**s and Human Review

Higher-**consequence** actions need narrower **permission**s and approval.



THINKER MISSION

Place **approval gate**s in a field-trip planning workflow.

Thinker name and date:

Write or draw your reasoning and evidence.

Readiness for the connected explanation

Recall Authentication and authorisation. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

Readiness for Privacy and least privilege

Does a duration summary need student names to calculate a total?

My readiness explanation

This week's end goal

By the end, I can enforce **least privilege** by checking operation and object before execution.

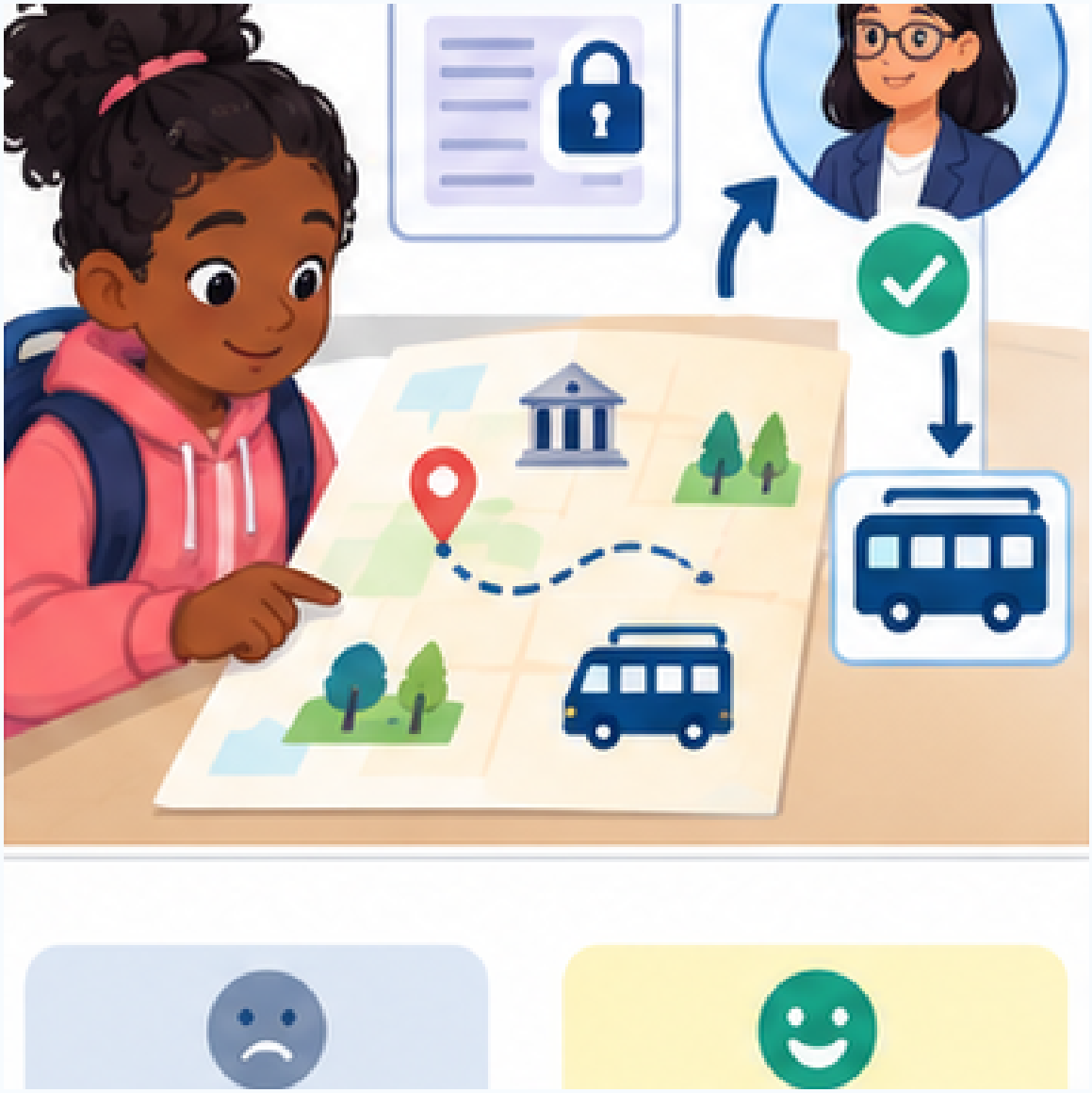
Notice the evidence

> Investigate the supplied Permissions and Human Review case. Separate observations, given facts and assumptions before choosing an explanation.

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



Start with this puzzle

MYSTERY

The learners must use **permission**s and human review to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

Write or draw your reasoning and evidence.

My **prediction** and the clue supporting it:

Write or draw your reasoning and evidence.

A closer explanation

Least privilege grants only capabilities required by the task. Tool access should include resource and operation scope, not merely a broad tool name.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Apply the connected idea: Privacy and least privilege

A fictional class activity dashboard needs category counts, not names or personal messages. Design its data and role **permission**s before collection.

This exercise uses invented data and policy cards; do not change the Academy's actual authentication settings.

STUDENT LESSON · 3 OF 18

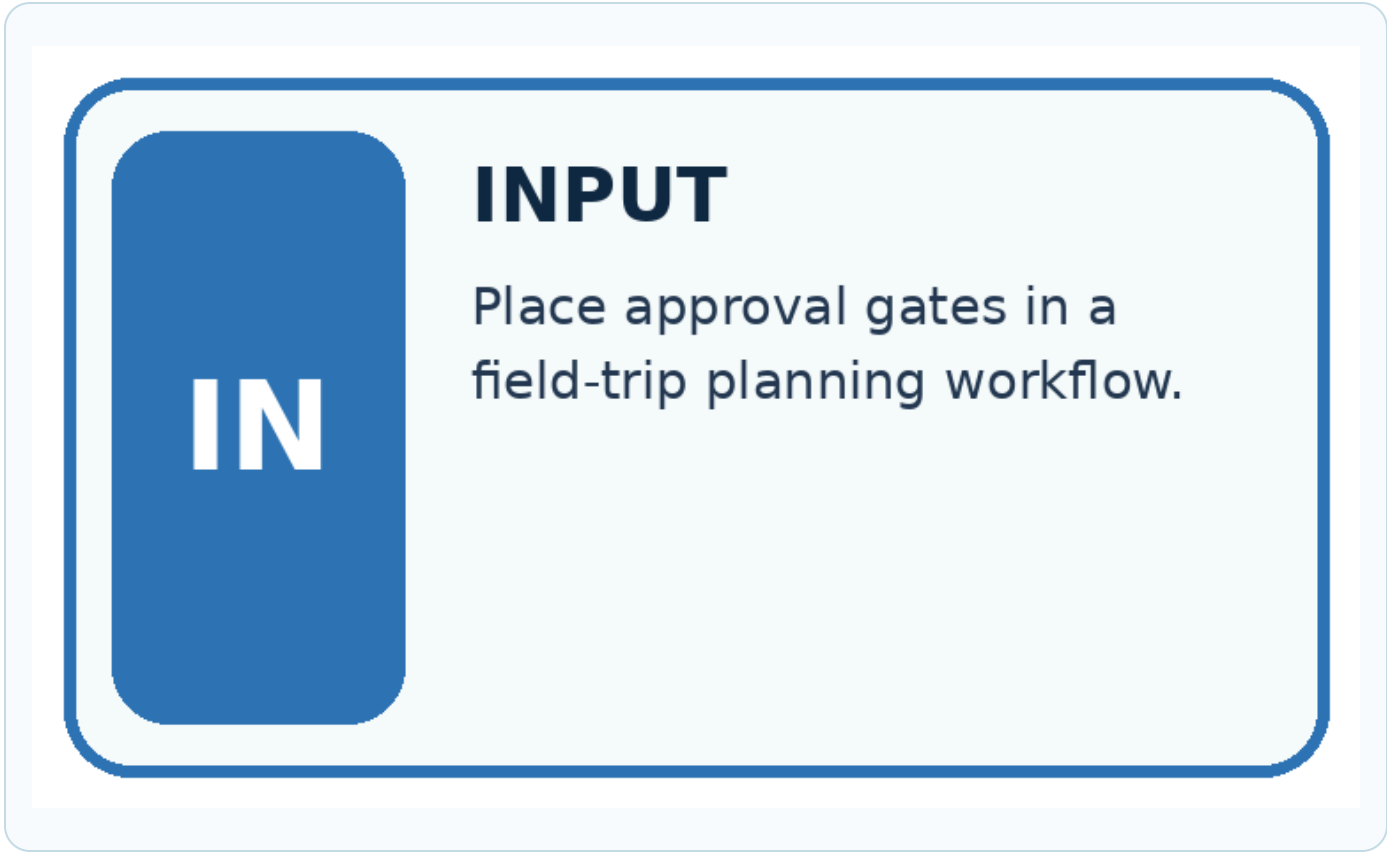
See how it works

> Follow the mechanism for Permissions and Human Review. Connect each labelled input, deciding step and result.

VISUAL MECHANISM

Input → **Process** → **Output**

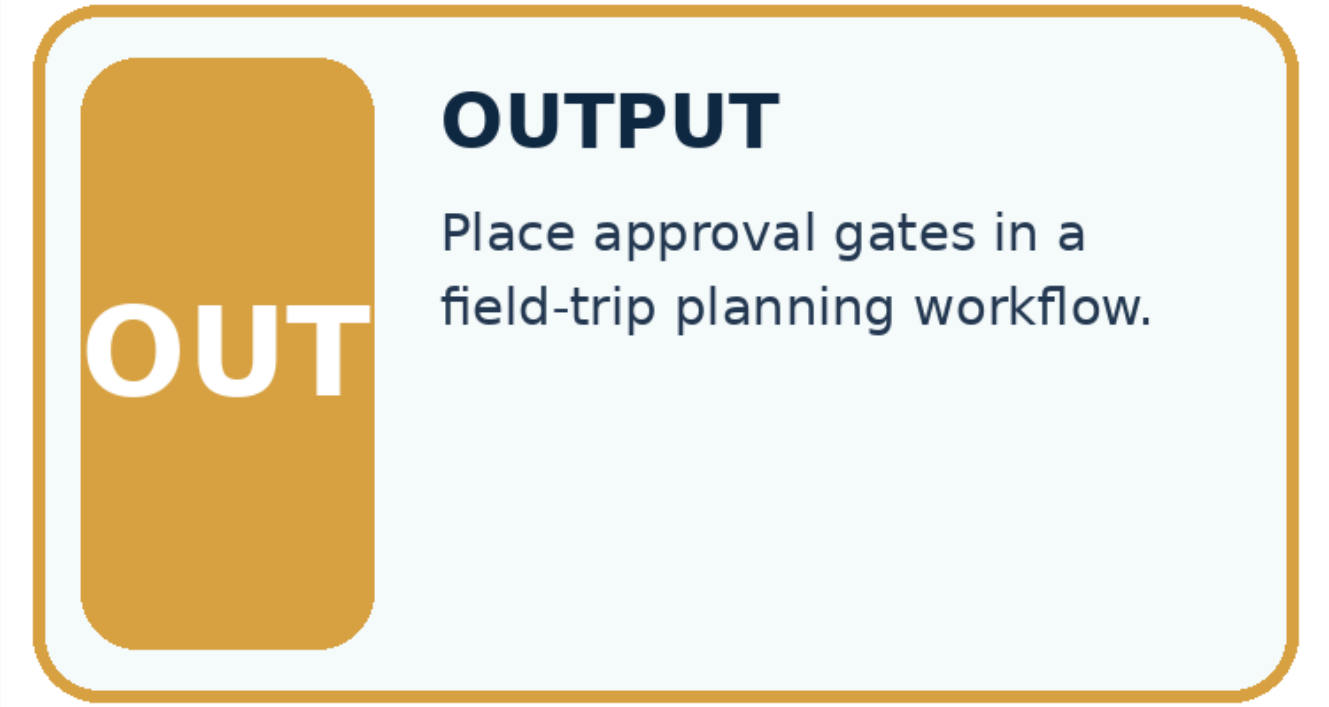
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Place approval gates in a field-trip planning workflow.

↓


PROCESS
Higher-consequence actions need narrower permissions and approval.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.

Least privilege and tool scopes — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
Tool token permits lookup(public-room). A proposed call writes teacher notes.	A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes. Check actor, operation and resource before executing a proposed call.	Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Apply the connected idea: Privacy and least privilege

```
roles = {"student": ["read_summary"],
         "teacher": ["read_summary", "read_review"]}
def allowed(role, action):
    return action in roles.get(role, [])

print(allowed("student", "read_summary"))
print(allowed("student", "read_review"))
print(allowed("unknown", "read_summary"))
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Student	Read summary	No review notes
Teacher	Read summary and review	Authorized purpose
Unknown role	No listed permissions	Deny by default

Read each row from left to right and connect the evidence to the stated rule.

The toy code prints True, False, False. It models a **permission** decision, not a complete security system.

A real server must verify identity and enforce **authorization** on every protected request; hiding a button in a page is insufficient.

Ask what each role actually needs

Least privilege means giving each role only the actions needed for its purpose. Read the policy as a table: students may read the approved summary, while teachers may also read the necessary review.

An unknown role gets no listed action. The small function checks this table; it does not establish who a visitor is.

In a real application, the server must verify the session and apply the **permission** check whenever protected data is requested.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

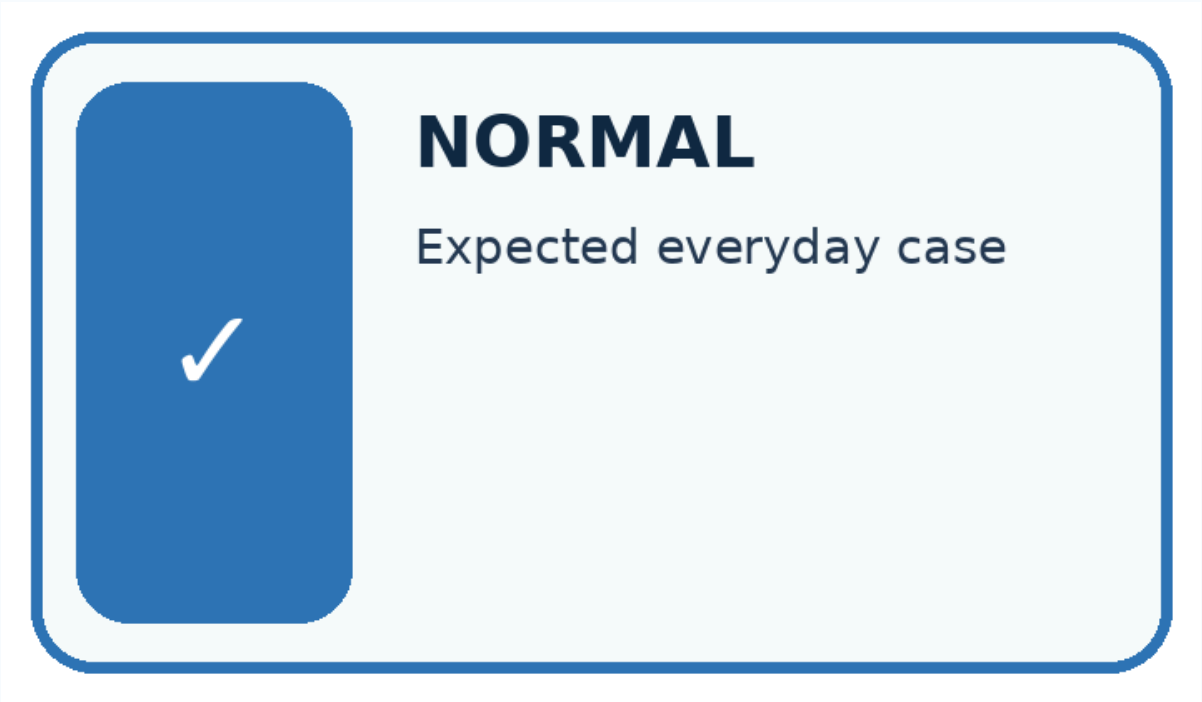
> Compare the supplied cases for Permissions and Human Review. Identify the detail that changes the conclusion.

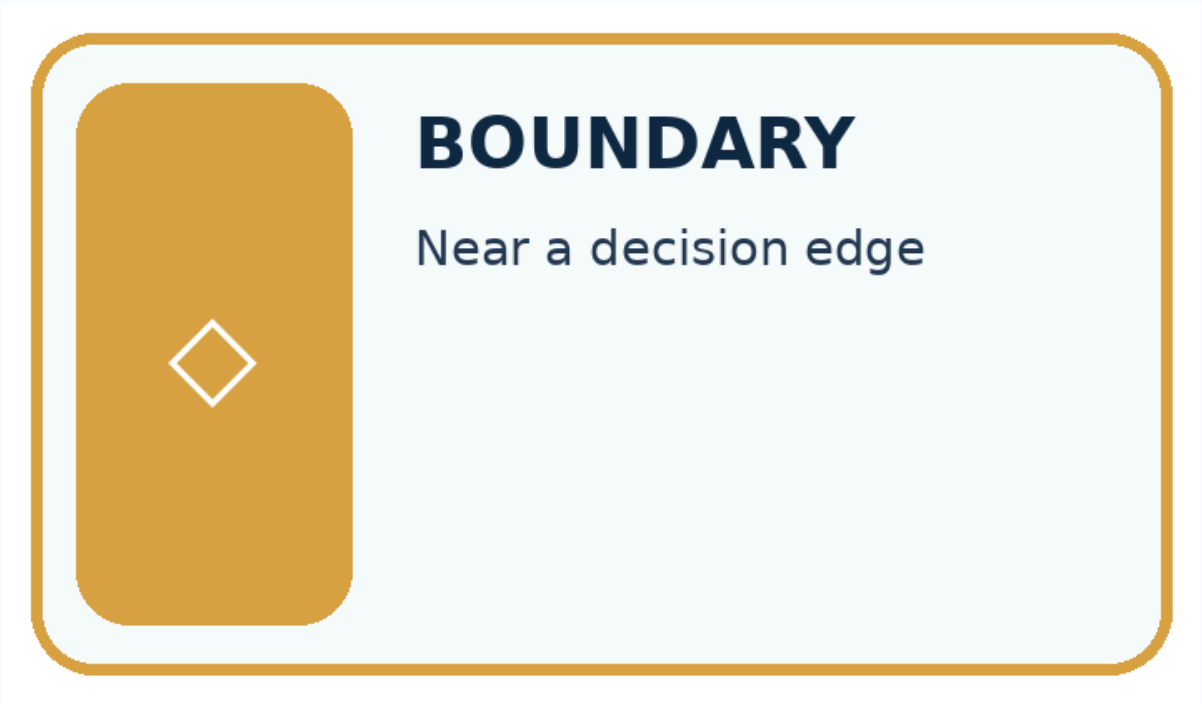
The case to investigate

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

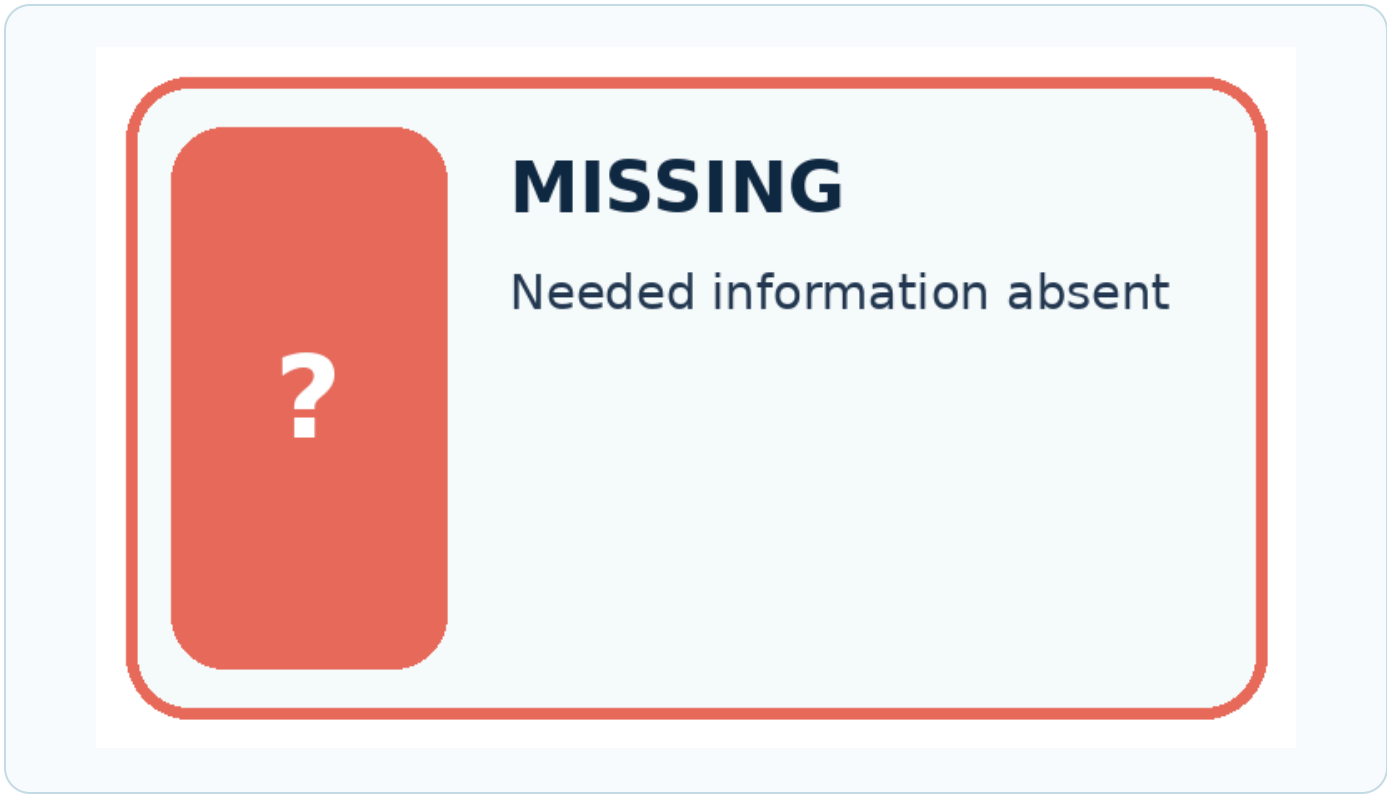
A system is not understood until it survives more than one comfortable example.





The normal case

NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.


?
MISSING
Needed information absent


→
TRANSFER
Fresh situation or user

Check missing information

MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Change one thing in the pictured case

Change one thing

Change: The helper says it needs write R5 to correct an apparent typo.

Predicted result

Prediction: The write remains denied under the existing token.

Why the result follows

Reason: A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

STUDENT LESSON · 5 OF 18

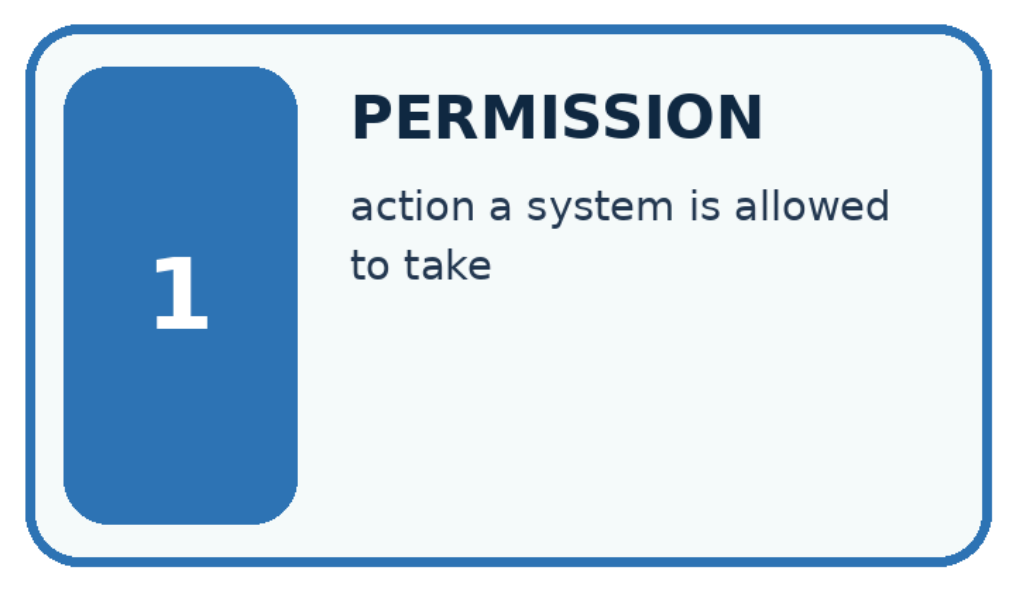
Words that unlock the idea

> Use the vocabulary of Permissions and Human Review to explain a concrete decision, rather than only memorising definitions.

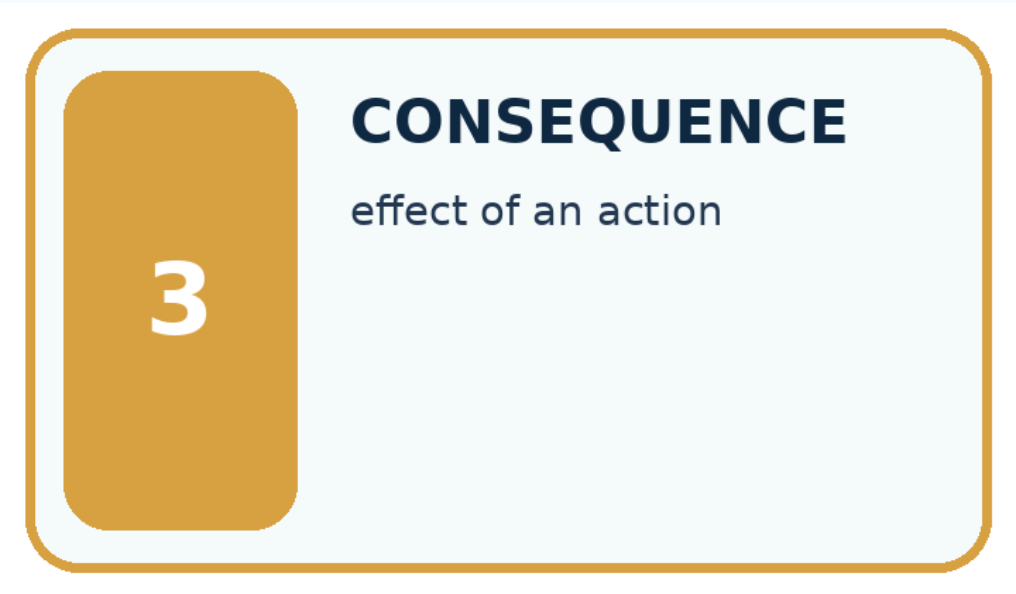
VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
APPROVAL GATE
pause requiring authorized human choice


3
CONSEQUENCE
effect of an action



Word	My own example	Not this / boundary
permission	Authorisation to perform a specified action on specified information or resources.	
approval gate	A point where a specified action must wait for explicit authorisation.	
consequence	An outcome or effect of a decision, including who benefits or is harmed.	
least privilege	Giving a user or system only the permissions needed for its authorised task.	

Words used in the closer explanation

Word	Meaning
Capability	A scoped permission to perform an operation
Resource scope	The specific records an operation may address

Terms for Privacy and least privilege

Term	Meaning
Data minimization	Keeping only data needed for the stated purpose
Least privilege	Giving a role only the access it needs
Retention	How long data is kept
Authorization	Permission to perform a particular action

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Permissions and Human Review, predict a step, then compare with its explanation.

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 31 • WORKED EVIDENCE

1. Low consequence: suggest three destinations.
2. Higher consequence: contact families, spend money, or submit forms.
3. The system drafts; an authorized adult reviews and decides.
4. Log what was approved, rejected, or changed, then stop after the approved scope.

1. Low **consequence**: suggest three destinations.

2. Higher **consequence**: contact families, spend money, or submit forms.

3. The system drafts; an authorized adult reviews and decides.

4. Log what was approved, rejected, or changed, then stop after the approved scope.

Recalculate, retrace, or restate the reasoning in your own representation:

Write or draw your reasoning and evidence.

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Tool token permits lookup(public-room). A proposed call writes teacher notes.

Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

Worked explanation · Permissions and Human Review

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Apply the connected idea: Privacy and least privilege

If a student manually requests the review endpoint, should hiding its menu link be enough?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Permissions and Human Review

A fictional catalogue helper receives a narrowly scoped tool token.

The facts we will use

Supplied facts

Token allows read on catalogue records R2 and R5 only. No write **permission** is granted.

Supplied facts

Proposals: read R2; read R8; write R5. Generated text claims all are helpful.

AI ACADEMY / AI-2/31

Permissions and Human Review

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Read token

Authorised operation is read; authorised objects are R2,R5.

2. Allowed call

read R2 matches both operation and object: allow.

3. Wrong object

read R8 fails object scope: deny.

4. Wrong operation

write R5 fails operation scope: deny even for an allowed record.

CHANGE ONE THING

The helper says it needs write R5 to correct an apparent typo.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Read token	Authorised operation is read; authorised objects are R2,R5.
2. Allowed call	read R2 matches both operation and object: allow.
3. Wrong object	read R8 fails object scope: deny.
4. Wrong operation	write R5 fails operation scope: deny even for an allowed record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Permissions are concrete restrictions, not a confidence score or a general instruction to be useful. Each proposed call must match both dimensions before it happens.

A request written inside catalogue text cannot enlarge the token’s authority. Denial should preserve the proposed action and reason without executing the disallowed mutation.

What this example does not establish

Keep the stated limit

Permission checks do not establish that a read result is trustworthy or that an allowed action is factually correct.

Keep the stated limit

Read the labelled picture: Which proposal is allowed, and why is an authorised object alone insufficient?

Keep the stated limit

Change one thing: The helper says it needs write R5 to correct an apparent typo. Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Permissions and Human Review claim and explain a repair using the supplied evidence.

Find the first error

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: If an action is convenient, the system should receive permission automatically. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Check the claim against the facts

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.

Apply the connected idea: Privacy and least privilege

Retention must be tied to a purpose and school policy rather than a guessed universal number of days. Identify who can correct or delete records and avoid keeping raw responses indefinitely just because storage is available.

Does the allowed function alone prove an app is secure?

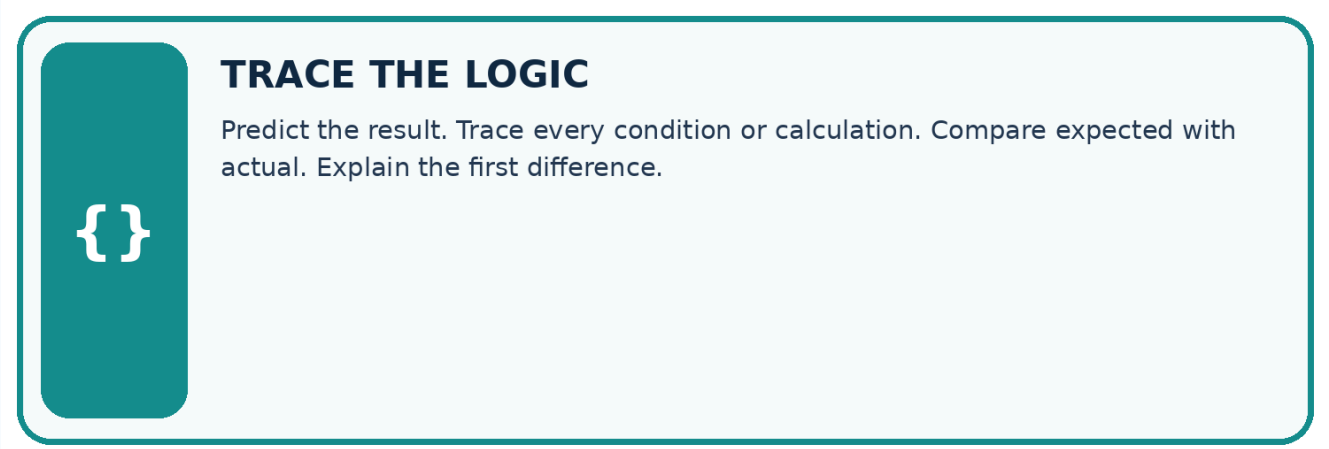
Practise together

> Practise Permissions and Human Review with support, then explain the deciding step in your own words.

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

> Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

A generated plan cannot expand the token or actor scope.

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

> **Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

Apply the connected idea: Privacy and least privilege

A form asks for names, birthdays and comments to count activity preferences. Remove fields not needed for the purpose.

Aggregated counts can still expose information in very small groups; choose reporting rules with the responsible teacher.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “**Least privilege** and tool scopes”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for **Least privilege** and tool scopes. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Privacy and least privilege

Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

> Which proposal is allowed, and why is an authorised object alone insufficient?

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Permissions and Human Review. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Map actions by **consequence**, grant minimum **permission**s, add adult gates, test a denied request, and verify the workflow stops safely.

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

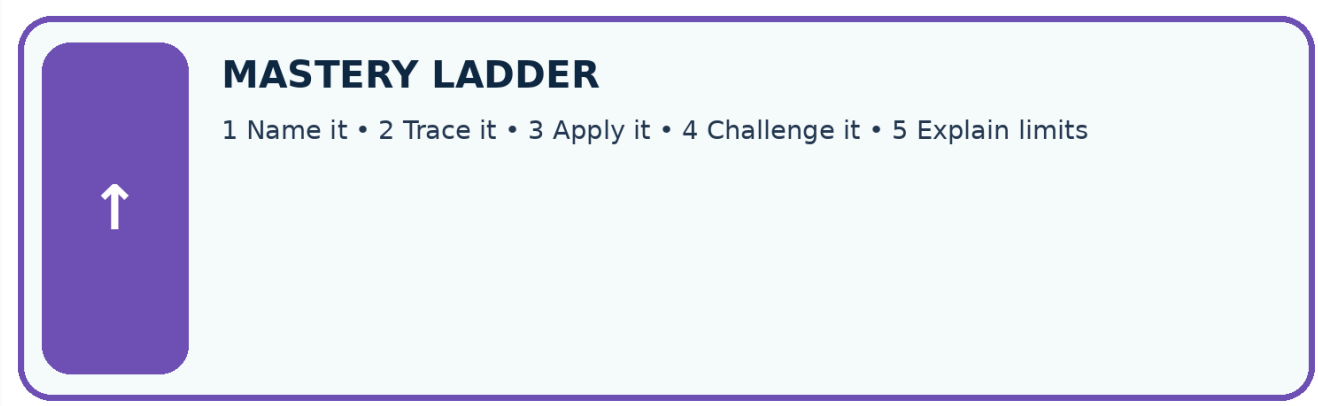
Build your understanding

> Move from recall to tracing, diagnosis and transfer in Permissions and Human Review. Explain why each step is justified.

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



1 • NAME: Define the concept without repeating the textbook.

2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.

3 • APPLY: Solve a different example independently.

4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

My strongest **evidence** of independent understanding:

Write or draw your reasoning and evidence.

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Transparent Book Recommender. Keep an annotated decision and the evidence supporting it.

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT The recommender may suggest and explain; people decide, borrow, purchase, or share data.

Trace the system

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

Responsible design

RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

Evidence for your project

Give the recommender read-only access wherever sufficient, and require explicit owner review for catalogue edits or publication.

STUDENT LESSON · 12 OF 18

Show what you know

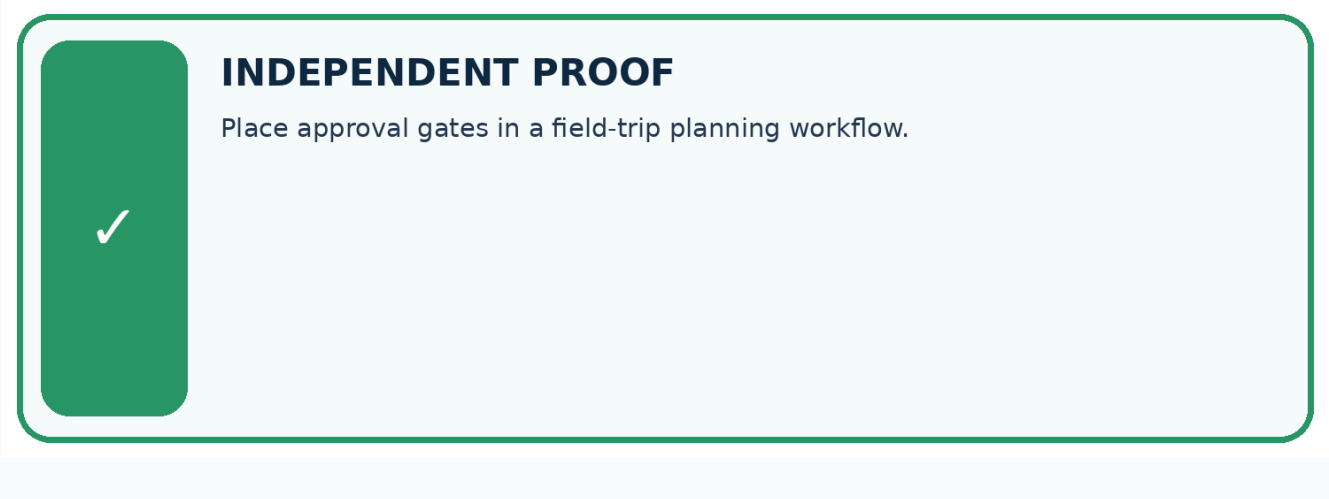
> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Independent evidence

INDEPENDENT MASTERY PROOF

Explain • Transfer • Test • Defend

Complete this page without copying the worked example. Your reasoning matters more than decoration.



Try a more demanding case

CHALLENGE Place **approval gate**s in a field-trip planning workflow.

1. My fresh example, trace, calculation, or design:

Write or draw your reasoning and evidence.

2. My **evidence** that it works (include a boundary or missing-data case):

Write or draw your reasoning and evidence.

3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Check your explanation

SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

STUDENT LESSON · 13 OF 18

Study a complete case

> Revisit the complete Permissions and Human Review case for supported practice. Repeated examples do not become fresh assessments.

Week 31 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Posting a recommendation requires teacher review.

The question

Place the **approval gate**.

Worked explanation

After draft and **evidence** are visible, before posting; include content and destination. No approval means no post.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Put approval after evidence and before action

Evidence or stage	Value or result
Draft	Show recommendation content
Evidence	Show support and destination
Teacher	Review and explicitly approve
Post	Proceed only with approval

Posting a recommendation requires teacher review.

Place the **approval gate**.

After draft and **evidence** are visible, before posting; include content and destination. No approval means no post.

STUDENT LESSON · 14 OF 18

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Scope allows read inventory items A, B. Calls read A, read C, write B. Decide.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 31 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A teacher allows a tool to draft a message but forbids sending. The draft is correct. May the agent send it? Explain the boundary.

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

The recommender may suggest and explain; people decide, borrow, purchase, or share data.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Transfer practice from the original programme

A teacher approves posting version 1 to a class board. The assistant changes the text and destination to a public board. Does the old approval cover the new action? Define what must be reviewed.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Permissions and Human Review, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: A fictional catalogue helper receives a narrowly scoped tool token.

By the end, I can	How I will show it
Explain the mechanism and terminology of least privilege and tool scopes	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Permissions and Human Review without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of **least privilege** and tool scopes

> **My deciding step, evidence and one remaining question**
>
> _____
>
> _____
>
> _____
>
> _____

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional catalogue helper receives a narrowly scoped tool token.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? write R5 fails operation scope: deny even for an allowed record.

> 1. Wrong operation

2. Read token

3. Wrong object

4. Allowed call

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Wrong operation write R5 fails operation scope: deny even for an allowed record.

Permissions are concrete restrictions, not a confidence score or a general instruction to be useful. Each proposed call must match both dimensions before it happens.

A request written inside catalogue text cannot enlarge the token’s authority. Denial should preserve the proposed action and reason without executing the disallowed mutation.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The helper says it needs write R5 to correct an apparent typo. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. The write remains denied under the existing token.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: The write remains denied under the existing token. A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

Repair a missing building block

Review Agents and Tools (ai-2/30). Goal: Explain the **mechanism** and terminology of state machines for bounded agents. Short practice: Can the assistant accept Trace B because its final state is done?.

Return to this week and explain what you changed.

Review Prompt Experiments (ai-2/28). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Name the case P passes and Q fails, and explain why it matters..

Return to this week and explain what you changed.

Review Text as Pieces (ai-2/25). Goal: Encode and decode text under an explicit toy vocabulary. Short practice: Why is the unknown ! at position 10 rather than position 9?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

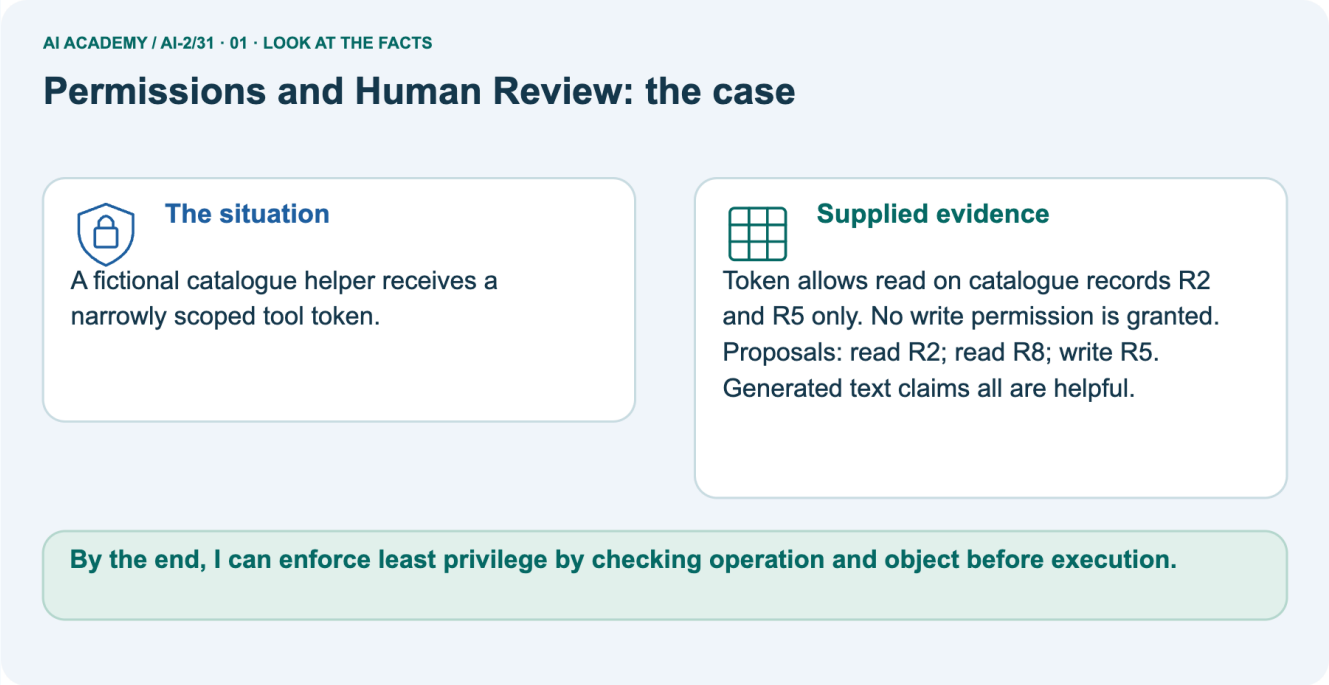
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Permissions and Human Review graphic using its labels, deciding evidence and relevant limitation.

01. Permissions and Human Review: the case



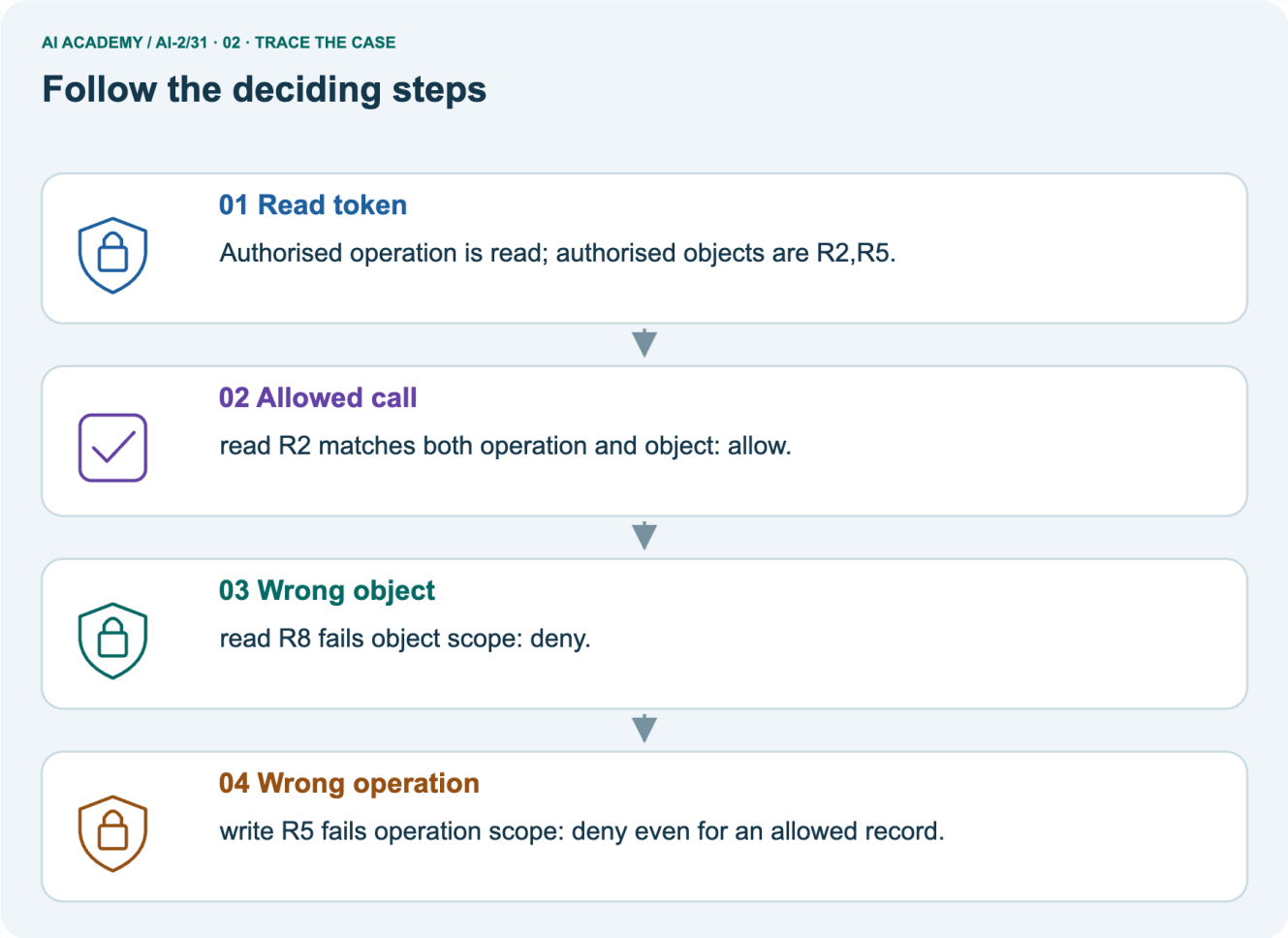
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional catalogue helper receives a narrowly scoped tool token. Supplied facts: Token allows read on catalogue records R2 and R5 only. No write **permission** is granted. Proposals: read R2; read R8; write R5.

Generated text claims all are helpful.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

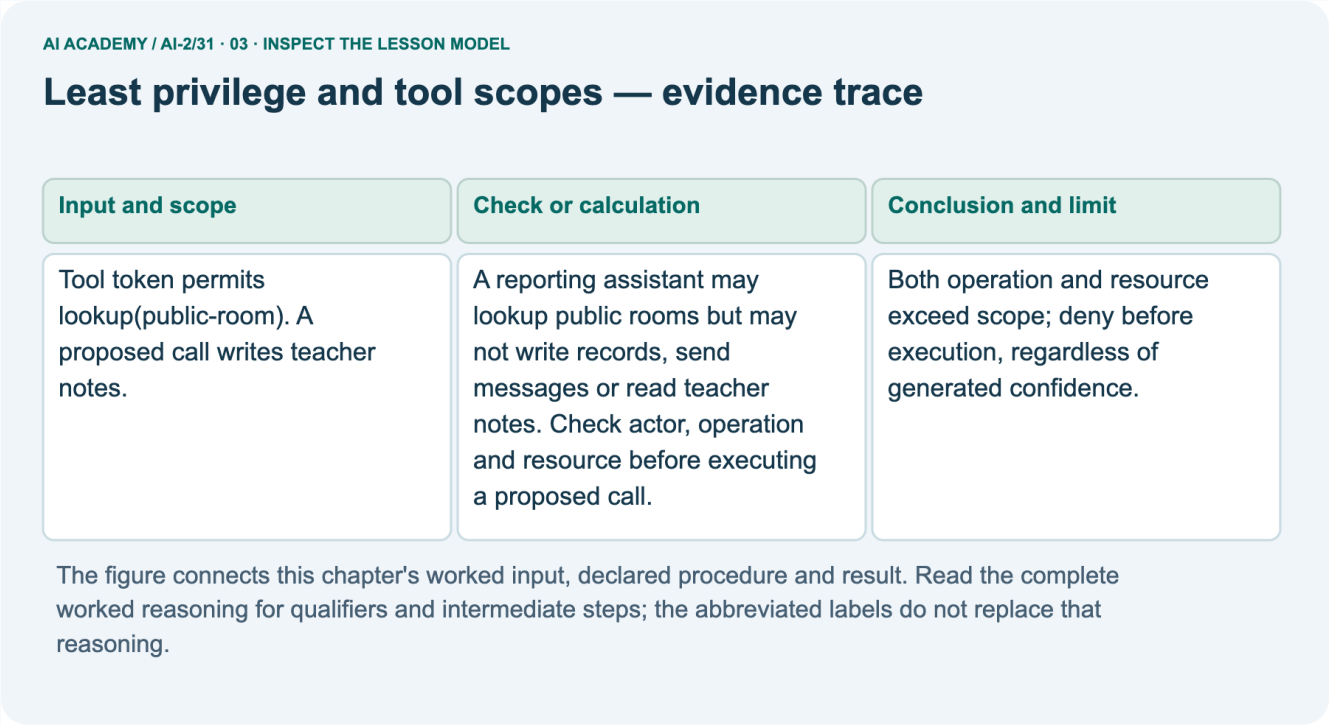


Follow the labelled picture with the matching explanation. Open the vector diagram

Read token: Authorised operation is read; authorised objects are R2,R5. → Allowed call: read R2 matches both operation and object: allow. → Wrong object: read R8 fails object scope: deny. → Wrong operation: write R5 fails operation scope: deny even for an allowed record.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Least privilege and tool scopes — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

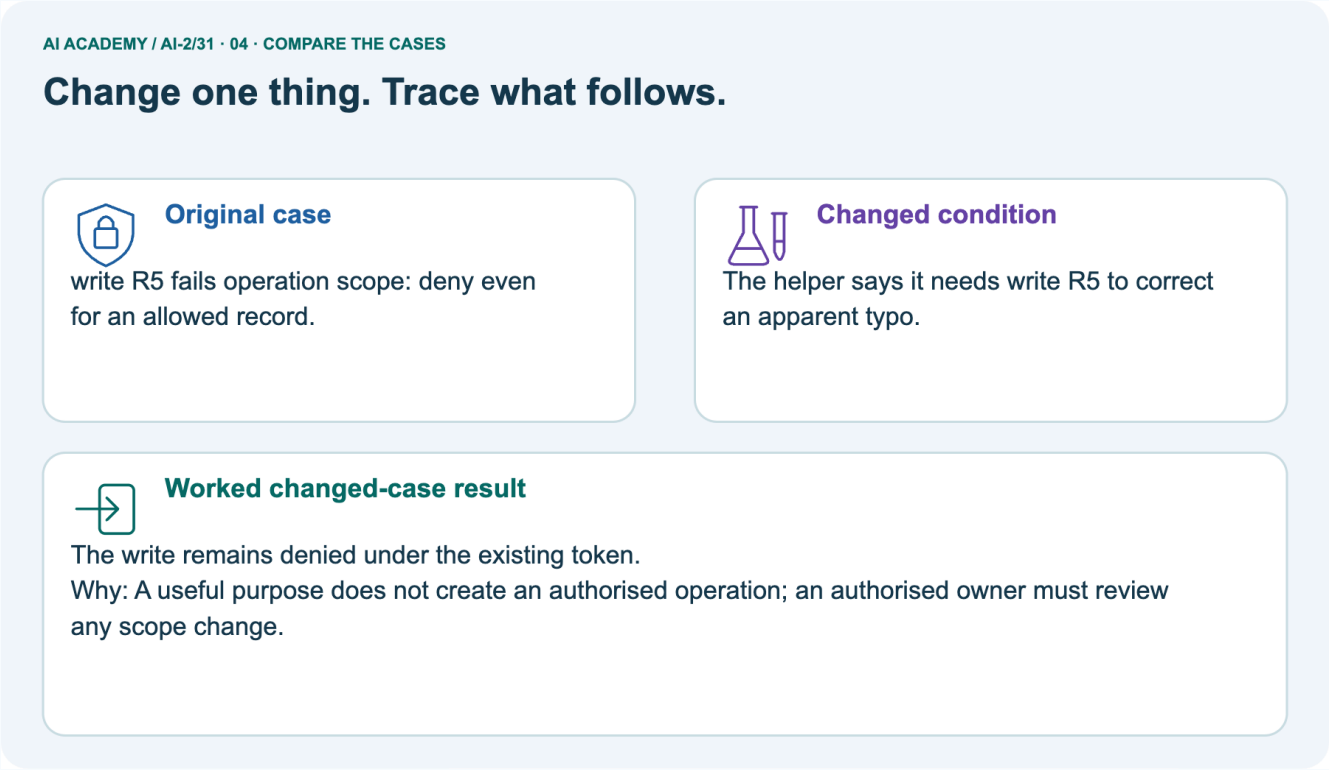
Least privilege and tool scopes — **evidence** trace. Tool token permits lookup(public-room).

A proposed call writes teacher notes. | A reporting assistant may lookup public rooms but may not write records, send messages or read teacher notes.

Check actor, operation and resource before executing a proposed call. | Both operation and resource exceed scope; deny before execution, regardless of generated confidence.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



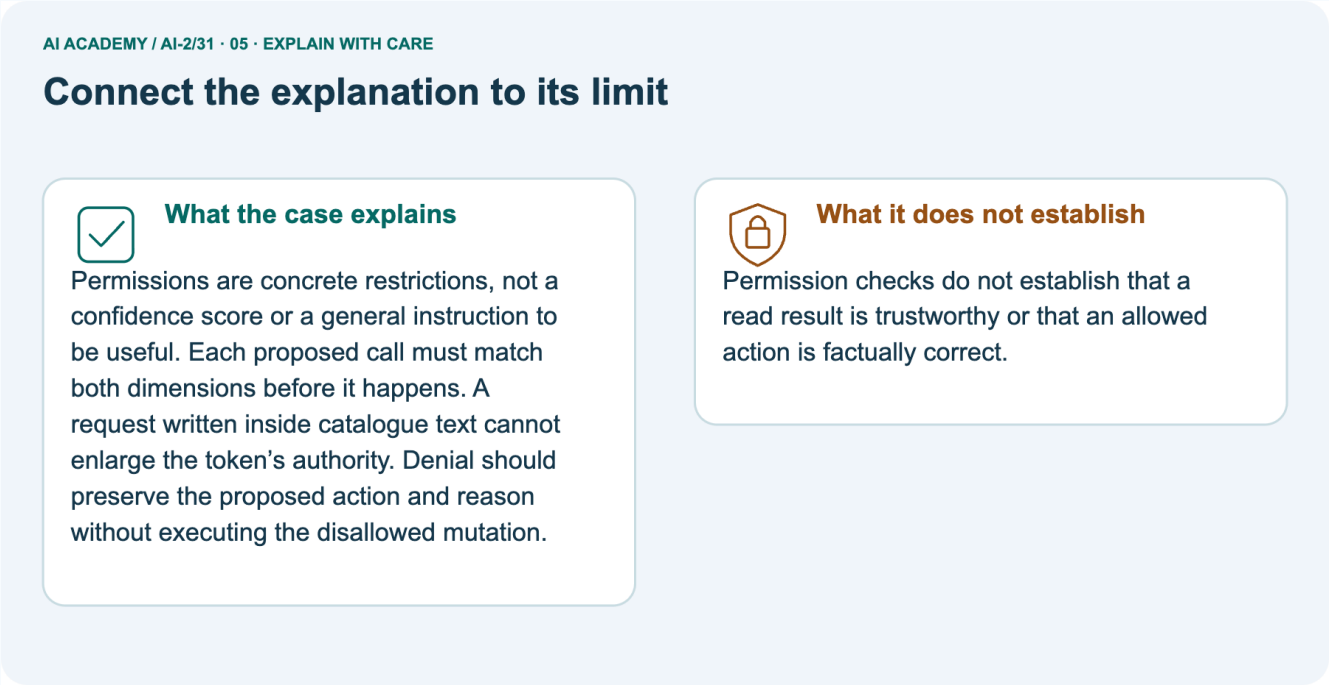
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: write R5 fails operation scope: deny even for an allowed record. Changed condition: The helper says it needs write R5 to correct an apparent typo.

Expected result: The write remains denied under the existing token. Reason: A useful purpose does not create an authorised operation; an authorised owner must review any scope change.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

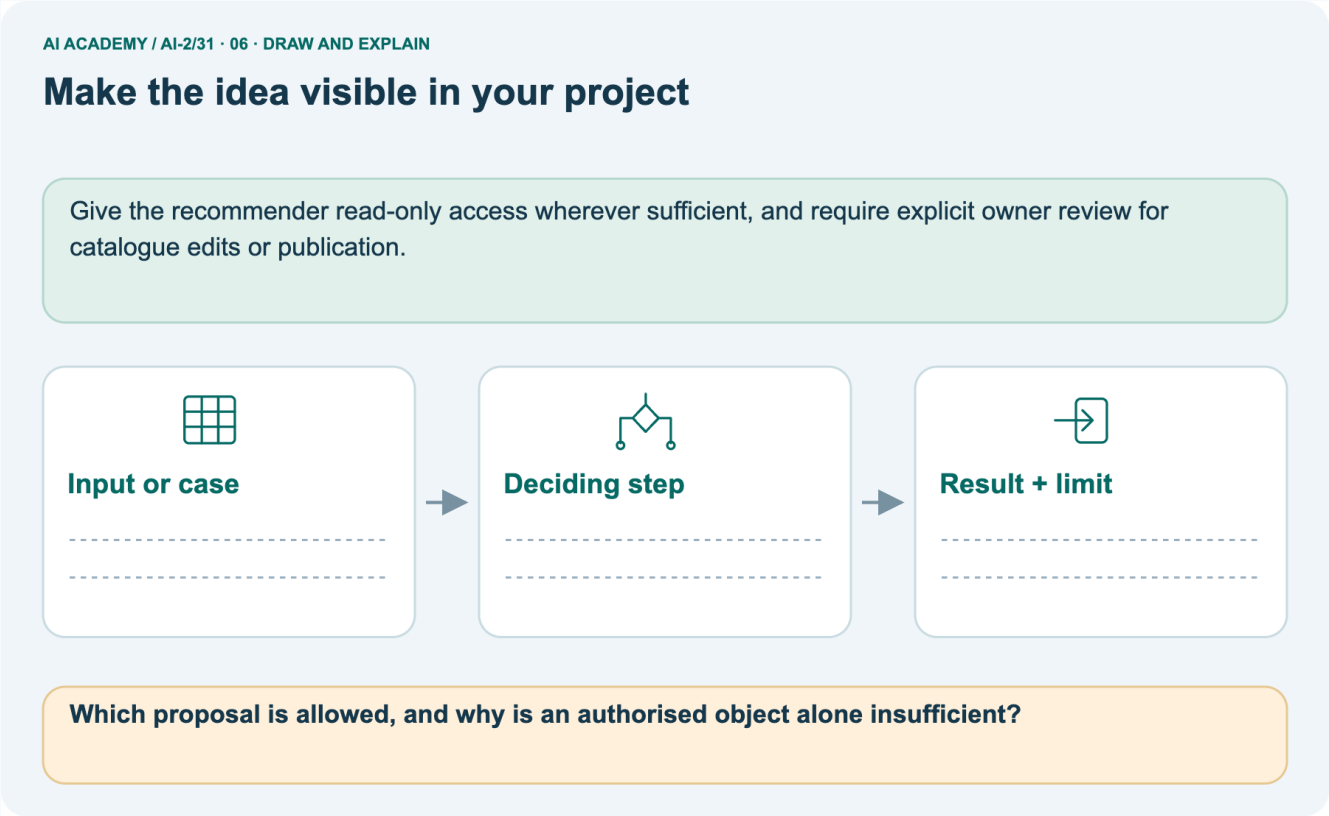
Permissions are concrete restrictions, not a confidence score or a general instruction to be useful. Each proposed call must match both dimensions before it happens.

A request written inside catalogue text cannot enlarge the token's authority. Denial should preserve the proposed action and reason without executing the disallowed mutation.

Limit: **Permission** checks do not establish that a read result is trustworthy or that an allowed action is factually correct.

> Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Give the recommender read-only access wherever sufficient, and require explicit owner review for catalogue edits or publication. Draw your **input**, deciding step, result and limit.

Supported question: Which proposal is allowed, and why is an authorised object alone insufficient?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.